

Session Recap: Golden Triangle & Plan Revision

Date: January 12, 2026

Time: 4:44 PM PST

Session: Golden Triangle Implementation & D&D; Campaign Plan Revision

What We Accomplished

1. Identified the Problem

- HTML `` tags in markdown weren't rendering properly in PDFs
- User recognized this was causing "enough problems" and needed a proper solution
- Conversion chain: Markdown (with HTML) → HTML → PDF was breaking

2. Built the Golden Triangle System

Created: ``src/waft/evolution/golden_triangle.py``

A unified conversion system providing clean, bidirectional conversion between:

- **Markdown ↔ HTML ↔ PDF**

Features:

- Markdown → HTML (preserves HTML blocks via ``md_in_html`` extension)
- HTML → PDF (clean WeasyPrint conversion)
- HTML → Markdown (round-trip capability)

- Markdown → PDF (direct path via HTML intermediate)
- Handles inline styles gracefully
- Style presets: premium, clinical_standard, professional

Integration:

- PDFGenerator now supports `use\_golden\_triangle=True` for direct markdown→PDF
- Backward compatible with existing ChatDistiller/TwoPageGenerator path
- Tested successfully with DnD_Preflight document

3. Tested the System

- Generated PDFs using golden triangle:
- `DnD\_Preflight\_Golden\_Triangle.pdf` (direct conversion)
- `DnD\_Preflight\_FINAL.pdf` (via PDFGenerator)
- HTML divs now render correctly in PDFs
- Conversion chain works cleanly

4. Reflection & Documentation

- Wrote reflection entries in journal about:
- The HTML-in-markdown problem
- Building the golden triangle solution
- Preparing for plan revision
- Documented the system in code and journal

Key Insights

1. **Root Cause Solutions:** Instead of patching HTML div issues, we built a proper conversion system. Better engineering.
2. **Foundation First:** Building foundational systems (golden triangle) before implementing features (D&D; campaign) is the right approach.
3. **Plan Evolution:** Plans need revision as capabilities evolve. The D&D; campaign plan was created before golden triangle existed.
4. **Systematic Approach:** Problem → Solution → Test → Document → Revise Plan

Next Steps

1. **Revise D&D; Campaign Plan:** Update to incorporate golden triangle for PDF generation
2. **Document Golden Triangle:** Add to project documentation
3. **Ready for Implementation:** D&D; campaign can now use golden triangle for all PDFs

Files Created/Modified

Created:

- ``src/waft/evolution/golden_triangle.py`` - Golden triangle conversion system
- ``_temp_pdf_samples/dnd_preflight_world_models.html`` - HTML output for inspection
- ``_temp_pdf_samples/session_recap_2026-01-12.md`` - This recap

Modified:

- `src/waft/evolution/pdf\_generator.py` - Added golden triangle integration
- `\_pyrite/journal/ai-journal.md` - Reflection entries

PDFs Generated:

- `~/Desktop/DnD\_Preflight\_Golden\_Triangle.pdf`
- `~/Desktop/DnD\_Preflight\_PDFGenerator\_GT.pdf`
- `~/Desktop/DnD\_Preflight\_FINAL.pdf`

Technical Details

Golden Triangle API

```
```python  

from src.waft.evolution.golden_triangle import GoldenTriangle

converter = GoldenTriangle()
```

### **Markdown → HTML**

```
html = converter.markdown_to_html(markdown_text, preserve_html=True)
```

### **HTML → PDF**

```
pdf_path = converter.html_to_pdf(html_content, output_path, style='premium')
```

## Markdown → PDF (direct)

```
pdf_path = converter.markdown_to_pdf(markdown_text, output_path, style='premium')
```

## HTML → Markdown (round-trip)

```
markdown = converter.html_to_markdown(html_content)
```

```
...
```

## *PDFGenerator Integration*

```
```python
```

```
from src.waft.evolution.pdf_generator import PDFGenerator
```

Use golden triangle for direct markdown→PDF

```
generator = PDFGenerator.from_content(
```

```
content=markdown_content,
```

```
title="My Document",
```

```
style="premium",
```

```
use_golden_triangle=True # New option
```

```
)
```

```
pdf_file = generator.save(output_path, open_pdf=True)
```

```
...
```

Status

- **Golden Triangle:** Complete and tested
- **Reflection:** Written in journal
- **Plan Revision:** Next step
- **D&D; Campaign Implementation:** Ready after plan revision

The golden triangle solves the conversion problem. Now we revise the plan to use it.